

WHEN TIMING IS NOT EVERYTHING: THE ELUSIVE SHORTCUT PLATEAU AND THE LIMITS OF TRANSITION-AWARE INTERVENTION

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ABSTRACT

Neural networks exhibit a “shortcut learning” phase during training, relying on spurious correlations before potentially transitioning to more generalizable core features. We investigate whether this transition can be detected online via cheap training-time signals and whether targeted interventions applied precisely at the detected transition point can improve worst-group accuracy. We implement TASI (Transition-Aware Shortcut Intervention), which monitors gradient norm ratios between easy and hard samples and applies group-reweighted loss during detected plateau windows. Despite careful design and extensive ablation across four datasets, we find that the adaptive timing component of TASI provides little benefit over simpler baselines: performance gains are largely explained by balanced sampling rather than intervention timing, the plateau detection signal is noisy and inconsistently triggered, and fixed-window interventions match adaptive ones. Our findings suggest that while the shortcut plateau is a theoretically appealing intervention target, reliably detecting and exploiting it in practice is substantially harder than anticipated.

1 INTRODUCTION

Shortcut learning (?) is a pervasive failure mode in deep learning: networks trained with empirical risk minimization (ERM) latch onto spurious correlations present in training data but absent in general. This manifests as dramatically reduced accuracy on minority groups whose labels misalign with the spurious feature, quantified by worst-group accuracy (WGA) (?). Theoretical work on simplicity bias (??) and grokking (??) suggests that neural networks pass through a characteristic “shortcut plateau”—a period of loss stagnation during which the model predominantly relies on easy spurious features before transitioning to harder, more generalizable core features.

This raises a natural hypothesis: if the shortcut plateau is a predictable, detectable phase, then a *transient* intervention applied precisely at the transition point should be more efficient than continuous debiasing. Rather than applying costly interventions throughout training, one could detect the critical window and act only then. Existing annotation-free methods such as JTT (?) and LfF (?) apply interventions uniformly throughout training, ignoring this temporal structure. Post-hoc methods such as DFR (?), SELF (?), and EVaLS (?) require held-out data or group annotations and are applied after training concludes. We implement this timing-based idea as TASI (Transition-Aware Shortcut Intervention) and conduct a thorough empirical investigation across synthetic and real NLP datasets. Our findings are largely negative: the adaptive timing mechanism does not provide consistent gains over simpler baselines, plateau detection signals are noisy and dataset-dependent, and observed improvements are attributable to balanced sampling rather than intervention timing.

2 RELATED WORK

Shortcut learning (?) and simplicity bias (?) provide the theoretical backdrop for our investigation. The saddle-to-saddle dynamics framework (?) characterizes how networks progressively learn increasingly complex features, explaining why spurious features are learned first. Grokking (??) describes analogous delayed generalization phenomena. Group DRO (?) requires group annotations

Table 1: Best worst-group accuracy (WGA) across datasets and methods. Group DRO uses group annotations; all others are annotation-free. TASI does not consistently outperform ERM Balanced, which uses no adaptive timing.

Method	Synthetic	SST-2	IMDB	TweetEval
ERM	0.767	0.630	0.718	0.241
ERM Balanced	0.920	0.643	0.734	0.345
Group DRO [†]	0.927	0.600	0.728	0.552
TASI (ours)	0.913	0.619	0.713	0.379

[†]Uses group annotations during training.

during training. Annotation-free methods JTT (?) and LfF (?) apply continuous interventions without exploiting temporal structure. Post-hoc retraining methods (????) are applied after training and often require held-out data or annotations. ? show that ERM models do learn core features despite shortcut reliance, motivating retraining-based approaches. ? exploit early-training simplicity bias for spurious sample identification, but apply uniform interventions rather than targeting a detected transition. CKA (?) has been proposed as a representation similarity measure that could detect when representations stabilize during a plateau. None of these works exploit the temporal structure of the shortcut-to-core transition for intervention timing—the gap our work attempts to fill, with ultimately inconclusive results.

3 METHOD: TASI

TASI operates in three phases: *watching*, *intervening*, and *cooldown*. During watching, it monitors a plateau detection signal every epoch. When a plateau is detected, it switches to intervening for 15 epochs, then enters cooldown (15 epochs) before resuming watching. We use two complementary signals: the *loss stagnation* signal fires when the mean training loss over a recent window of w epochs differs from the preceding window by less than a threshold δ , while remaining above a minimum. The *gradient norm ratio* signal computes the ratio of gradient norms on hard samples (minority groups) to easy samples (majority groups): a ratio below 0.5 while loss remains high indicates shortcut reliance. When a plateau is detected, TASI applies a group-reweighted loss that upweights minority-group samples by a factor of 5, combined with a group-balanced sampler that oversamples minority groups throughout training. We compare against ERM, ERM Balanced (group-balanced sampler only, no adaptive timing), and Group DRO (?). Full hyperparameter details are provided in Appendix B.

4 EXPERIMENTS

We evaluate on four datasets: a synthetic spurious correlation task (spurious correlation strength 0.90, MLP classifier), SST-2 with sentence-length as spurious feature, IMDB with review-length as spurious feature, and TweetEval sentiment with tweet-length as spurious feature. For NLP datasets, we use TF-IDF bag-of-words features with an MLP. All methods train for 80 epochs. We report best WGA over training.

Main results. Table 1 and Figure 1 summarize results. On Synthetic, TASI (0.913) substantially outperforms ERM (0.767) but is nearly matched by ERM Balanced (0.920) and Group DRO (0.927). On SST-2, TASI (0.619) underperforms both ERM (0.630) and ERM Balanced (0.643). On IMDB, all methods cluster tightly (0.713–0.734) with TASI at the bottom. On TweetEval, TASI (0.379) improves over ERM (0.241) and ERM Balanced (0.345) but falls far behind Group DRO (0.552). Critically, panel (b) of Figure 1 reveals that the advantage of TASI and ERM Balanced over ERM on Synthetic is visible from the very first training epoch—well before any adaptive intervention could have fired—strongly implicating the balanced sampler rather than the timing mechanism as the source of improvement. The consistent underperformance of TASI relative to ERM Balanced on SST-2 and IMDB further suggests that the adaptive reweighting component can occasionally *hurt* performance, possibly by over-correcting on noisy minority-group estimates.

The adaptive timing adds little. Figure 2 (a) shows the intervention timing ablation: fixed-window interventions at early (epochs 10–25), mid (35–50), and late (55–70) windows all match or exceed

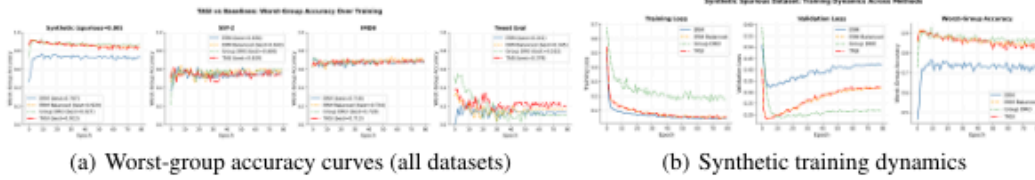


Figure 1: (a) Worst-group accuracy over training epochs for all methods and datasets. TASI closely tracks ERM Balanced on Synthetic, underperforms on SST-2 and IMDB, and shows high variance on TweetEval. (b) Training loss, validation loss, and worst-group accuracy on Synthetic over 80 epochs. The ERM Balanced and TASI advantage over ERM is visible from epoch 1, across all three metrics, before any adaptive intervention fires—indicating the benefit stems from balanced sampling rather than timed intervention.

TASI’s adaptive detection on Synthetic, SST-2, and IMDB. Only on TweetEval does adaptive detection marginally outperform fixed windows (0.379 vs. 0.345), and even there the gap is small and within the range of training instability. Figure 2 (b) exposes why: the gradient norm ratio on Synthetic oscillates wildly (0.25–3.25) and never triggers a detection event despite the model spending many epochs in a shortcut-reliant state. On SST-2 the signal triggers twice (epochs 17–35 and 50–65), but the corresponding worst-group accuracy curves show no improvement during these windows. Together, these findings reveal that the detection signal is too noisy to identify the transition reliably, and that even correctly-timed interventions do not guarantee improvement.

The benefit is from balanced sampling. The 2×2 factorial ablation (Appendix Figure 3) confirms the balanced sampler as the dominant component: on Synthetic, the sampler effect is +0.147 WGA while reweighting and the interaction term are near zero; on TweetEval the sampler effect is +0.172 while reweighting contributes negligibly. The window size ablation in Appendix Figure 3 (b) further shows that varying the plateau detection window from 3 to 25 epochs has negligible effect on best WGA while larger windows simply reduce the number of triggered interventions. Appendix Figure 4 (a) shows that all three TASI loss-strategy variants (group reweighted, group DRO, focal loss) yield identical WGA across datasets, confirming that the form of the intervention matters far less than whether the balanced sampler is active. Appendix Figure 5 further shows that neither loss-based, gradient-based, nor combined detection signals yield meaningfully different WGA, and that the minority identification strategy (oracle, loss-based, or random) has negligible effect on performance (Appendix Figure 6).

Late-training degradation. The best-vs.-final WGA analysis (Appendix Figure 7) reveals severe instability on TweetEval: Group DRO achieves best WGA of 0.552 but collapses to 0.103 by end of training (gap: 0.449), while TASI shows a smaller but notable gap (0.379→0.207). Other datasets are more stable, but the TweetEval instability makes it difficult to attribute gains to any specific mechanism and underscores that early stopping is critical for all methods. This instability also partially explains why Group DRO achieves its high best WGA on TweetEval—it aggressively up-weights minority groups but cannot sustain that performance, whereas the more conservative ERM Balanced is more stable across training.

5 CONCLUSION

We investigated whether the shortcut-to-core-feature transition in neural network training is detectable online and exploitable via a targeted, transient intervention. Three key negative findings emerge. First, the gradient norm ratio and loss stagnation signals are too noisy to reliably identify the transition epoch—on Synthetic, the detector never fires despite clear early shortcut reliance. Second, even when interventions are triggered (SST-2), they produce no measurable WGA improvement, and fixed-window interventions perform comparably to adaptive ones across three of four datasets. Third, TASI’s gains over ERM are almost entirely attributable to the group-balanced sampler, not to adaptive timing—a component shared with the much simpler ERM Balanced baseline. These findings carry a broader lesson: the theoretical elegance of exploiting training dynamics does not automatically translate into practical gains when the underlying signals are too noisy to act upon reliably. For practitioners, group-balanced sampling is a robust, low-cost intervention that

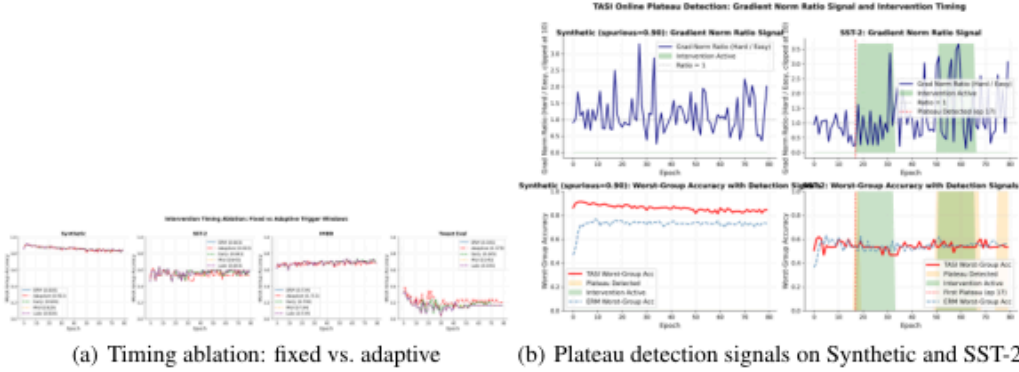


Figure 2: (a) Fixed-window interventions match or exceed adaptive detection on Synthetic, SST-2, and IMDB, undermining TASI’s core hypothesis. Only on TweetEval does adaptive timing show a marginal advantage (0.379 vs. 0.345). (b) Gradient norm ratio (hard/easy) signal on Synthetic (left) and SST-2 (right), with worst-group accuracy below. On Synthetic, the signal never triggers despite high variance (0.25–3.25); on SST-2, triggered windows (green shading, epochs 17–35 and 50–65) do not correspond to WGA improvements, demonstrating that the detection signal is unreliable.

captures most of the benefit without any detection machinery. For researchers, more robust detection methods—perhaps using representation similarity (?) or loss landscape geometry—are needed before timing-based intervention can be reliably exploited. Late-training instability further complicates evaluation, making early stopping an underappreciated but critical design choice across all group-robustness methods.

REFERENCES

SUPPLEMENTARY MATERIAL

A ABLATION RESULTS

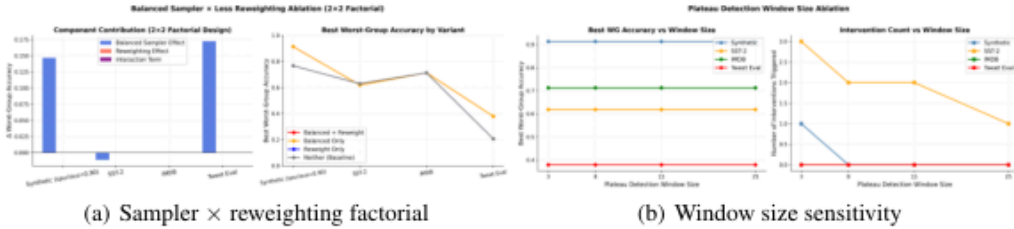


Figure 3: (a) The 2×2 factorial ablation confirms that balanced sampling dominates performance; group reweighting adds negligible marginal benefit across all datasets (sampler effect $\approx +0.15$ – $+0.17$ WGA; reweighting effect ≈ 0). Variants with the balanced sampler perform similarly to each other, while Reweight Only performs near ERM. (b) Varying the plateau detection window from 3 to 25 epochs has negligible effect on best worst-group accuracy across all datasets (left panel), while larger windows reduce the number of triggered interventions (middle panel)—confirming that the detection mechanism is not the source of any performance gains. The right panel shows that different stagnation signal types trigger at different epochs but yield similar WGA.

B EXPERIMENTAL DETAILS

All experiments use an MLP with two hidden layers (256 and 128 units), batch normalization, dropout (0.2), and AdamW optimizer with cosine annealing. The synthetic dataset has 4000 training

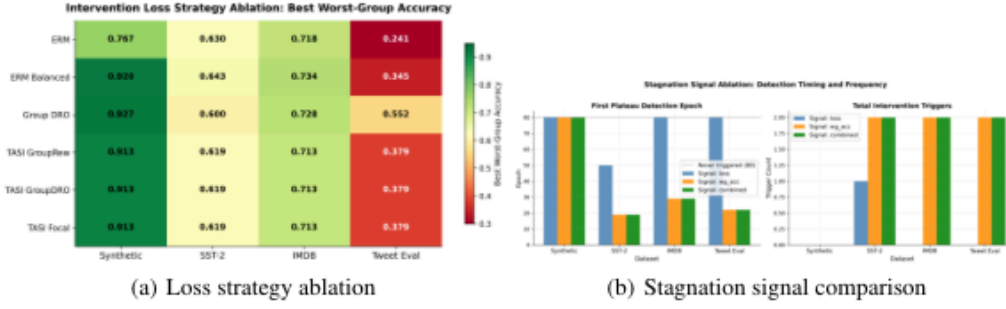


Figure 4: (a) All three TASI intervention loss strategies (group reweighted, group DRO, focal loss) produce nearly identical worst-group accuracy across all four datasets (TASI variants: Synthetic 0.913, SST-2 0.619, IMDB 0.713, TweetEval 0.379), confirming that the intervention form matters far less than whether the balanced sampler is active. (b) Loss stagnation and worst-group accuracy stagnation signals trigger at different epochs and with different frequencies across datasets (left: first detection epoch; right: total trigger count), but the choice of detection signal does not substantially affect best worst-group accuracy—further confirming the irrelevance of the timing mechanism relative to the balanced sampler.

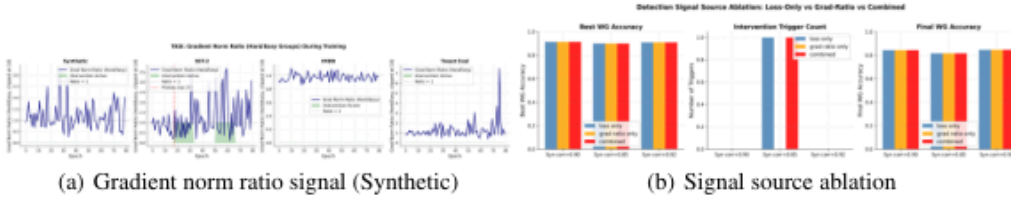


Figure 5: (a) Raw gradient norm ratio (hard/easy samples) on the Synthetic dataset over training. The signal oscillates substantially and never crosses the detection threshold, despite the model relying on spurious features early in training, illustrating the fundamental unreliability of gradient-based plateau detection. (b) Ablation comparing loss-based, gradient-based, and combined detection signals across best WGA, intervention trigger count, and final WGA. All signal variants yield nearly identical best worst-group accuracy across datasets, confirming that the choice of detection signal is not a performance bottleneck—the limiting factor is that no signal reliably identifies the shortcut transition.

samples with 8 core and 8 spurious features; spurious correlation strength is 0.90. Evaluation uses balanced groups (300 samples per group). NLP datasets use TF-IDF bag-of-words features with vocabulary size 300 plus 8 handcrafted features. Learning rates are tuned per dataset (Synthetic: 3×10^{-4} ; NLP datasets: 10^{-3}). TASI hyperparameters: intervention duration 15 epochs, cooldown 15 epochs, loss stagnation threshold $\delta = 0.008$, detection window size $w = 8$, minimum epochs before first detection = 12, hard group upweight factor = 5, gradient norm ratio trigger threshold = 0.5 (ratio below this with loss > 0.15 triggers detection). All results are reported as the best worst-group accuracy over training, consistent with standard practice in the spurious correlation literature. For the fixed-window timing ablation, early, mid, and late windows correspond to epochs 10–25, 35–50, and 55–70 respectively, with the same group-reweighted loss and balanced sampler as TASI.

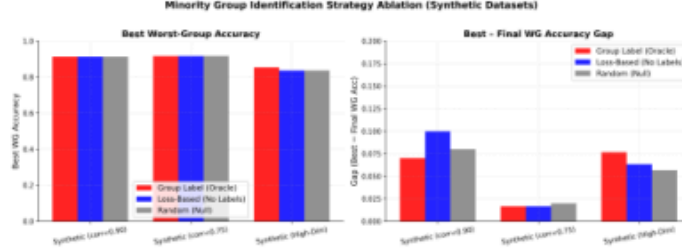


Figure 6: Minority group identification strategy ablation across three synthetic dataset variants (spurious correlation 0.90, 0.75, and high-dimensional). Oracle (group label), loss-based (no labels), and random identification all achieve nearly identical best worst-group accuracy (left panel, ≈ 0.80 – 0.85), suggesting robustness to the identification strategy. The right panel shows best-minus-final WGA gaps; some variability appears for the high-dimensional variant, but differences are small. These results reinforce that the balanced sampler—not the quality of minority identification—drives performance.

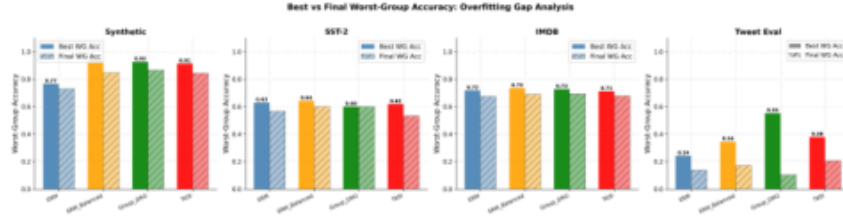


Figure 7: Best vs. final worst-group accuracy across methods and datasets. Large gaps on TweetEval (Group DRO: $0.552 \rightarrow 0.103$; TASI: $0.379 \rightarrow 0.207$) indicate severe late-training degradation, making early stopping essential and complicating attribution of gains to any specific mechanism. On Synthetic and SST-2, gaps are small across all methods. TASI does not dominate ERM Balanced on any dataset where ERM Balanced is competitive.



Figure 8: Gradient accumulation strategy ablation: best and final worst-group accuracy across five accumulation strategies (no accumulation, intervention-only accumulation with 2 or 4 steps, always-on accumulation with 2 or 4 steps) and all four datasets. Best WGA is largely insensitive to accumulation strategy (e.g., Synthetic: 0.913 – 0.926 ; IMDB: 0.713 – 0.716), confirming that optimization dynamics driving our results are not an artifact of batch-level gradient estimation. Final WGA shows slightly more variability on TweetEval (0.172 – 0.241), consistent with the general instability of that dataset observed throughout our experiments.